DEVESH SRIVASTAVA

EDUCATION

Indian Institute of Technology (BHU), Varanasi *2024 - 2028*
Bachelor of Technology in Chemical Engineering *8.79*

Mary Lucas School *2023*
CICSE *90.50%*

Mary Lucas School *2021*
CICSE *92.00%*

SKILLS

- **Programming Languages & Subjects:** C/C++, Python, Java, JavaScript, TypeScript, SQL; OS, OOP, DBMS
- **Technologies:** React, Node.js, Express, PostgreSQL, Prisma, MongoDB, Supabase, Socket.IO, Docker
- **Areas of Interest:** Data Structures, Algorithms, Competitive Programming, Machine Learning, Web Development
- **Machine Learning & Optimization:** PyTorch, Scikit-Learn, TensorFlow, XGBoost, Pandas, NumPy, Pyomo, HiGHS

PROJECTS

MarsSenitel - Unsupervised Image Screening *GitHub*
Python, PyTorch, ConvAE, Isolation Forest, Gaussian Mixture Models

- Built a five-stage convolutional autoencoder with a **256-D** representation for **10,422** Mars HiRISE crops; separated training, validation, and calibration by source observation.
- Recorded validation **SSIM 0.6738** and **MSE 0.002599**; used a **2,000-tree Isolation Forest** to rank latent-space novelty.
- Calibrated a BIC-selected Gaussian mixture threshold with **100 source-level bootstrap resamples**; flagged **17 screening candidates** from eight sources for inspection.

PharmaDash - Pharmacy Delivery Platform *Website | GitHub | Mar - Jun 2026*
TypeScript, Node.js, PostgreSQL, Prisma, Socket.IO *Solo Project*

- Built prescription verification, **FEFO inventory allocation**, five-role workflows, and order lifecycle management with **serializable stock transactions**.
- Implemented JWT/RBAC and object-level authorization, private S3 prescriptions with presigned URLs, and Razorpay payments with verified, idempotent webhooks.
- Added real-time rider GPS tracking and OTP delivery verification with brute-force protection; deployed using Docker and Render with **47+ integration tests**.

OptiGreen-Chem - Supply Chain Optimization *Website | GitHub*
XGBoost, PyTorch Geometric, Pyomo, HiGHS, Streamlit *Team Project: ML + Optimization*

- Developed an ML and optimization pipeline on **890 public EIA series** for a network of **12 refining districts and 58 regions**.
- Built P10/P50/P90 XGBoost forecasts across **229 state-product series**: P50 WAPE **4.40%** in 2018-19, **23.7% below the baseline**; also modeled refinery shortage risk.
- Formulated a rolling Pyomo/HiGHS MILP with integer marine cargoes, capacity constraints, and carbon pricing; **288 recorded plans** solved with gaps below **0.01%**.

POSITIONS OF RESPONSIBILITY

Avionics Head, Rocketry Division *Present*
Astronomy Club, IIT (BHU) Varanasi *Member since July 2025*

- Lead avionics work on telemetry, sensor integration, flight electronics, recovery systems, and the ground station for student-built rockets.
- Delivered PCB design workshops using **EasyEDA**; mentor junior members in avionics and hands-on system integration.

Design Executive, Society of Chemical Engineers

- Designed event posters and banners using Figma and Canva; contributed to branding for Osmoze'26, the society's annual carnival.

ACHIEVEMENTS

- **Second Place - NSSC 2026 Data Analytics, IIT Kharagpur:** MarsSenitel, 2-4 October 2026.
- **National Finalist - Impactify, Jagriti'25, IIT (BHU):** National case study competition.
- **Institute Finalist - Smart India Hackathon 2026:** AI-based waterborne disease prediction and surveillance.

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